Physics-Informed Neural Networks and Properties of Quantum Dot Systems

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STREAMLINE2 Collaboration Symposium, FRIB and MSU, May 21-22, 2026

Outline

Quantum Dots as Artificial Atoms

- Semiconductor quantum dots $\equiv$ artificial atoms
- Precise tunability: particle number N, confinement ω, interaction strength, external fields
- Intersection of condensed matter, quantum information and nanotechnology
- Rich many-body physics even for few particles:
  - Wigner molecule formation
  - Spin ordering
  - Non-trivial excitation spectra
- Clean realization of interacting fermions in reduced dimensions: Coulomb, exchange, quantum confinement all play essential roles

Key parameters

N	particle number
ω	trap frequency
λ	interaction strength

Core challenge

Hilbert space grows combinatorially with N

Acknowledgements

Many good friends and colleagues

Jane Kim (ANL), Patrick Cook (MSU), Danny Jammooa (MSU), Dean Lee (MSU), Ryan LaRose (MSU), Nick Cariello (MSU), Bryce Fore (ANL), Alessandro Lovato (ANL), Stefano Gandolfi (LANL), Alessandro Roggero (UniTN), Francesco Pederiva (UniTN), Arnau Rious (Barcelona), Giuseppe Carleo (EPFL), Niyaz Beysengulov (EEROQ), Johannes Pollanen (EEROQ, MSU), Zachary Stewart (MSU), Jared Weidman (MSU), Angela Wilson (MSU), Aleksander Sekkelsten (UiO), Francesco Massel (USN, UiO), Gunnar Lange (UiO), Viktor Svensson (UiO), Cecilie Glittum (UiO), Jonas Flaten (UiO), Oskar Leinonen (UiO), Øyvind Sigmundson Schøyen (UiO), Stian Dysthe Bilek (UiO), and Håkon Emil Kristiansen (UiO). Apologies to those omitted.

Review article

Alessandro Lovato, Giuseppe Carleo, Bryce Fore, MHJ, Jane Kim, Arnau Rios, Noemi Rocco, *Neural-network quantum states for the nuclear many-body problem*, Progress in Nuclear and Particle Physics, under review and `https://arxiv.org/abs/2602.13826`.

Why Existing Methods Struggle

Traditional methods and their limitations

- **Exact diagonalization:** restricted to very small N
- **Quantum Monte Carlo:** fermion sign problem; stochastic errors
- **DFT:** uncontrolled approximations; limited in strongly correlated regimes
- **Mean-field / perturbation:** fails when correlations are large

All methods struggle when exploring high-dimensional parameter spaces or modelling realistic experimental conditions.

Machine-learning approaches

1. Variational Monte Carlo (VMC) with neural ansatz
2. Physics-Informed Neural Networks (PINNs)

This talk

PINNs within a Slater–Jastrow–neural-correlator ansatz applied to quantum dots, $N = 2, 6, 12, 20$, across five orders of magnitude in confinement ω (oscillator frequency).

Why quantum dots? A long digression: Di Vincenzo Criteria for Quantum Computing

Quantum computing requirements

1. A scalable physical system with well-characterized qubit
2. The ability to initialize the state of the qubits to a simple fiducial state
3. Long relevant quantum coherence times longer than the gate operation time
4. A universal set of quantum gates
5. A qubit-specific measurement capability

Our platform: electrons on helium (EEROQ/MSU)

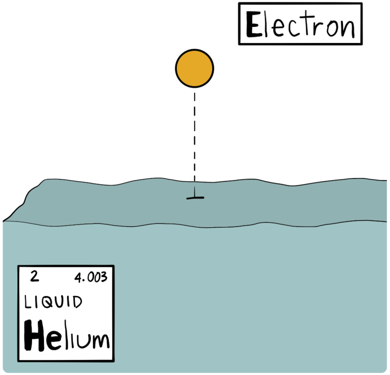
Key properties

1. Long coherence times
2. Highly connected qubits
3. Many qubits in a small area
4. CMOS compatible
5. Fast gates
6. G. Kolstra *et al.*, *High-impedance resonators for strong coupling to an electron on helium*, Phys. Rev. Applied **23**, 024001 (2025).

Single electrons can make great qubits

Electrons on helium

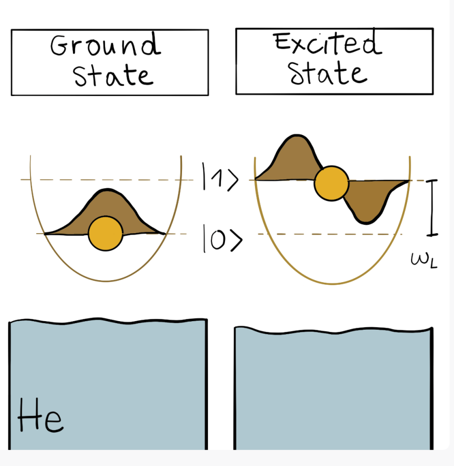
At the heart is the trapping and control of individual electrons floating above pools of superfluid helium. These electrons form the qubits of apotential paltform for quantum computer, and the purity of the superfluid helium protects the intrinsic quantum properties of each electron. The ultimate goal is to build a large-scale quantum computer based on quantum magnetic (spin) state of these trapped electrons.



Operations for quantum computing

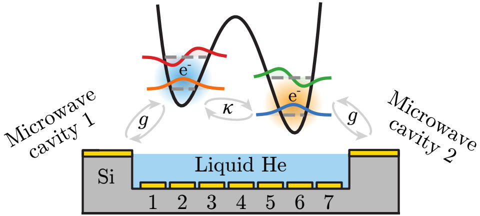
Electrons on helium

Quantum information can be encoded in a number of ways using single electrons. Currently, we are working with the side-to-side(lateral) quantum motion of the electron in the engineered trap. This motion can either be in its lowest energy state, the ground state, or in a number of higher-energy excited states. This electron motion also provides the readout capabilities for the ultimate goal of building a large-scale quantum computer based on the electron's magnetic moment (spin).



Final experimental and theoretical setup for Coulomb entanglement

1. Microdevice where two electrons are trapped in a double-well potential created by electrodes 1-7. The read-out is provided by two superconducting resonators dispersively coupled to electron's in-plane motional states.
2. Coupling constants from each individual electrode beneath the helium layer.
3. The electron's energy in a double-well electrostatic potential.
4. Screened Coulomb interaction



Recent work

- **Coulomb interaction-driven entanglement of electrons on helium**, Niyaz R. Beysengulov *et al.*, PRX Quantum **5**, 030324 (2024).
- **Design and Dynamics of High-Fidelity Two-Qubit Gates with Electrons on Helium**, Oskar Leinonen *et al.*, Physical Review A **113**, 032624 (2026)
- **Electrons on Helium and Entangled Quantum Sensors for Particle Physics**, Niyaz R. Beysengulov *et al.*, Communications Physics, in preparation

Two-Electron 2D Schrödinger via FCI with Slater Determinants

Key approach for identical fermions:

- We construct the two-electron basis using **Slater determinants** that include both spatial and spin degrees of freedom
- Each basis state is a properly antisymmetrized combination: $|\psi\rangle = |\phi_a\sigma_i; \phi_b\sigma_j\rangle$ where the total wavefunction is antisymmetric under particle exchange
- This naturally separates states into singlet (spatially symmetric) and triplet (spatially antisymmetric) configurations
- We include the full 2D soft Coulomb interaction and diagonalize the Hamiltonian in this antisymmetrized basis
- We compute both spin expectation values ($\langle S^2\rangle$, $\langle S_z\rangle$) and study the entanglement entropy for each eigenstate

Double-Well Trap Potential

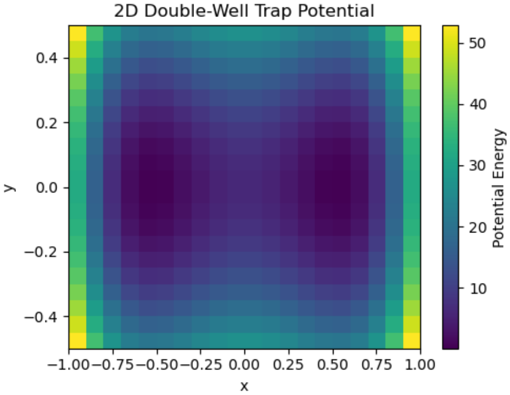
We use an arbitrary 2D potential $V(x, y)$ for the single electrons, meant to represent the electrostatic trapping potential from electrodes beneath the helium surface. For simplicity here we show an analytic double-well form:

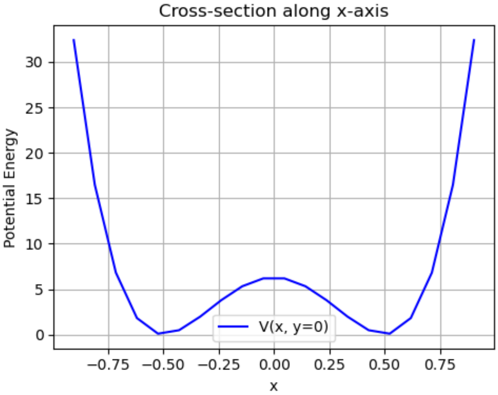
- Quartic double well in x: $V_x = k(x^2 - a^2)^2$
- Harmonic confinement in y: $V_y = \frac{1}{2}k_y y^2$

To allow specification of a 2D electrode geometry and voltages, one would solve the 3D Laplace equation for the electrode layout to find the coupling functions $\kappa_i(x, y)$ on the helium surface. Then the total trap potential is

$$V(x, y) = \sum_i \kappa_i(x, y)\, V_i$$

Harmonic oscillator double well, no anharmonic terms





Two-Electron Slater Determinant Basis for Identical Fermions

For **indistinguishable fermions**, we must construct properly antisymmetrized two-electron states using Slater determinants. Each basis state includes both spatial orbitals $\phi_i(\mathbf{r})$ and spin states $\sigma \in \{\uparrow, \downarrow\}$.
A two-electron Slater determinant is:

$$|\phi_a\sigma_i, \phi_b\sigma_j\rangle = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_a(\mathbf{r}_1)\sigma_i(1) & \phi_b(\mathbf{r}_1)\sigma_j(1) \\ \phi_a(\mathbf{r}_2)\sigma_i(2) & \phi_b(\mathbf{r}_2)\sigma_j(2) \end{vmatrix}$$

For each eigenstate in our Slater determinant basis, we compute: 1. **Spin expectation values**: $\langle S^2 \rangle$ and $\langle S_z \rangle$ 2. **Total entanglement entropy**: Via Schmidt decomposition including both spin and spatial degrees of freedom

Why compute total (spatial + spin) entanglement?

For **indistinguishable fermions**, the spatial and spin degrees of freedom are fundamentally coupled by antisymmetrization:

- **Singlet states**: Spatially symmetric $\times$ spin antisymmetric
- **Triplet states**: Spatially antisymmetric $\times$ spin symmetric

This coupling means that **pure spatial entanglement alone would often be zero** because:

- For a pure singlet in orbital —a,a⟩, there's only one spatial configuration (both electrons in the same orbital)
- The entanglement comes from the spin antisymmetry: $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$

The **total entanglement** (including both spin and spatial) captures the full quantum correlations between the two particles and is the physically meaningful measure for identical fermions. We use *Model-aware Reinforcement Learning* to optimize the various parameters in order to maximize entanglement.

What is Reinforcement Learning? (Used here to maximize entanglement)

Setting: electrons on helium

In our double-well setup, the **agent** controls the electrode voltages $\{V_i\}$; the **reward** is the total entanglement entropy of the two-electron state. RL finds the voltage configuration that maximizes quantum correlations.

General framework

The RL framework consists of an **agent** that learns how to solve a task by interacting with a physical system, called the environment. The agent can select **actions** that modify the **state** of the environment; as a consequence a **reward** is given to the agent, and the process repeats in a positive feedback loop. The objective is to maximize the sum of expected rewards and thereby accomplish the task. The agent learns a **strategy** that can be employed under similar yet different circumstances. The so-called **policy** is the rule for choosing actions.

Machine learning meets quantum hardware

Core challenge: Quantum devices are delicate systems. Small imperfections $\Rightarrow$ decoherence and loss of information.

Reinforcement learning has proven particularly powerful:

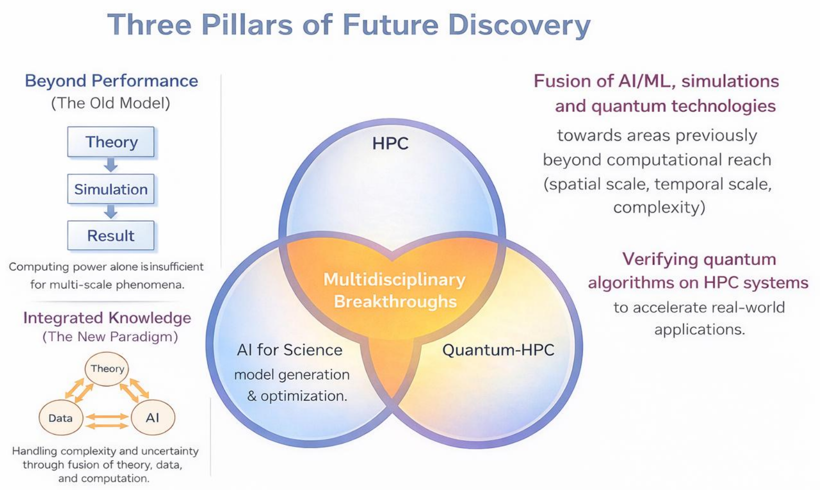
- Agent interacts with a simulated/experimental quantum system
- Learns control protocols that optimize a desired objective
- Applications: state preparation, high-fidelity gate implementation

ML Applications to Quantum Hardware

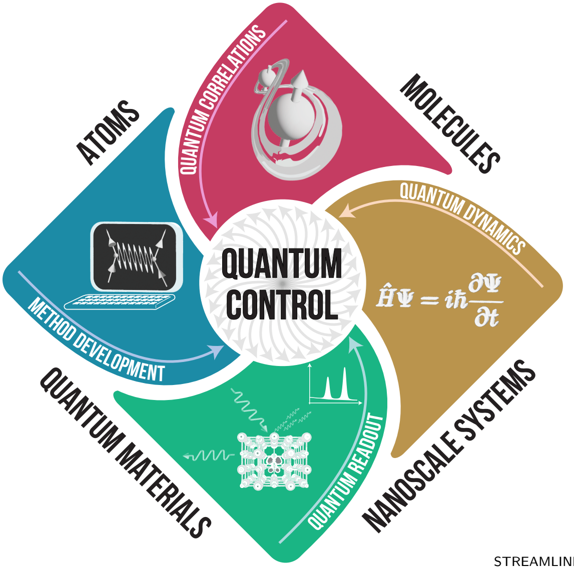
- Calibration & characterisation
- Error mitigation & noise reduction
- Automated control-sequence discovery
- Adaptive quantum experiment design

Parameter spaces in quantum devices are **huge** — traditional optimization struggles.

Conceptual Framework — Three Pillars of Future Discovery



Quantum Control



Neural Paradigms for Quantum Many-Body Problems

Method	**Core idea**	**Natural regime**
Variational MC (VMC) with neural ansatz	Minimise $\langle H \rangle$ via MC sampling	Large N, ground states, lattice systems
PINNs	Enforce PDE and symmetry residuals directly in loss	Continuous space, complex geometries, dynamics

Key PINN advantage for quantum dots

Wave function learned as a continuous function of particle coordinates; antisymmetry, normalization, and conservation laws imposed as explicit constraints or penalty terms.

PINN Formulation

Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ satisfy $\mathcal{N}[u](x) = 0$ on Ω with $u = g$ on $\partial\Omega$.

A PINN represents u by a neural network $u_\theta(x)$ and minimises:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \sum_i |u_\theta(x_i) - y_i|^2 + \lambda_{\text{PDE}} \sum_j |\mathcal{N}[u_\theta](\xi_j)|^2 + \lambda_{\text{BC}} \sum_k |u_\theta(\zeta_k) - g(\zeta_k)|^2$$

Data term

Observed data $\{x_i, y_i\}$

PDE residual

Collocation points ξ_j in Ω

BC term

Boundary pts ζ_k on $\partial\Omega$

Key tool — Automatic differentiation (AD): evaluates ∇u_θ and Δu_θ exactly (to floating-point precision), enabling end-to-end training constrained by physical laws.

Mitigation Strategies and Biases

Generalization in physics ML means consistency with physical laws, not merely accuracy on unseen data. Three interacting mechanisms:

1. Algorithmic stability of gradient-based optimization
2. Implicit bias toward smooth, low-complexity solutions
3. Architectural inductive biases encoding domain knowledge

Practical engineering checklist

- Weak/variational or energy formulations when strong form is unstable
- Hard-enforce essential boundary conditions
- Balance various loss terms
- Smooth activations; architectures controlling derivatives
- Exploit symmetry and equivariance
- Second-order optimizers (L-BFGS, Stochastic Reconfiguration)
- Importance/stratified sampling; space-time protocols

Does a PINN Respect the Variational Principle?

- Physics-Informed Neural Networks (PINNs) are increasingly used to solve quantum many-body problems.
- A central question:

Key question

Does a PINN approximation to the ground-state energy respect the variational principle?

- The answer depends crucially on how the PINN loss is formulated.
- Two strategies — variational and residual-based — have fundamentally different properties.

The Variational Principle

For a normalized trial wave function ψ_θ:

$$E[\psi_\theta] = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} \geq E_0$$

- E_0: exact ground-state energy
- Equality holds *only* for the exact ground state

Key property

Any admissible trial state gives an **upper bound** to the true ground-state energy.

Two PINN Training Strategies

A PINN represents the wave function as a neural network.

1. Variational (energy-based)

$$\mathcal{L}(\theta) = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle}$$

Equivalent to Rayleigh–Ritz minimization.
Variational principle preserved: $E_\theta \geq E_0$.

2. Residual minimization (PDE-based)

$$\mathcal{L}(\theta) = \left\| (H - E)\psi_\theta \right\|^2$$

Enforces the Schrödinger equation directly.
Residual minimization $\neq$ energy minimization.

Breakdown of the Variational Property

For residual-based PINNs there is **no guarantee** that:

$$E_\theta \geq E_0$$

Possible failure modes:

- Energies *below* the true ground state
- Convergence to incorrect eigenstates
- Residual small but solution error large (stability issues)

Conclusion

Residual-based PINNs do **not** preserve the variational principle.

Connection to PINN challenges

This is a manifestation of the *stability* challenge: small PDE residual $\not\Rightarrow$ small energy error.

Key Practical Issues

Normalization

- Must be enforced explicitly or via the energy quotient
- Unnormalized networks can give arbitrarily low apparent energies

Symmetry constraints

- Fermionic antisymmetry (Slater determinant)
- Spin and spatial symmetries

Nodal structure

- Especially critical for fermions
- Incorrect nodes $\Rightarrow$ incorrect ground state

Boundary conditions

- Hard enforcement preferred over soft penalties
- Soft boundary conditions can be dominated by other loss terms

Hybrid Loss Functions

A common approach combining the best of both strategies:

$$\mathcal{L}(\theta) = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} + \lambda \, \mathcal{L}_{\text{BC}} + \mu \, \mathcal{L}_{\text{sym}}$$

- Energy (Rayleigh–Ritz) term preserves variational character
- $\mathcal{L}_{\text{BC}}$: boundary-condition penalty
- $\mathcal{L}_{\text{sym}}$: symmetry (antisymmetry, spin) penalty

Recommendation

Avoid making the PDE residual the *dominant* loss term if variational upper bounds are required. Use the energy quotient as the primary objective.

Variational vs. Residual: Summary Comparison

Property	Variational PINN	Residual PINN
Loss function	$\langle\psi_\theta\|H\|\psi_\theta\rangle/\\|\psi_\theta\\|^2$	$\\|(H-E)\psi_\theta\\|^2$
Variational bound	Yes: $E_\theta \geq E_0$	No guarantee
Enforces PDE	Implicitly (at minimum)	Explicitly
Normalization	Automatic via quotient	Must be added
Risk of sub-E_0	None	Present
Connection to QMC	Direct (Rayleigh–Ritz)	Indirect

Variational Structure: Takeaways

When variational principle is preserved

- Loss $\equiv$ energy Rayleigh quotient
- Normalization enforced via quotient
- Antisymmetry exact (Slater determinant)
- SR / natural-gradient optimization tail

When it is not

- Dominant PDE residual loss
- Soft normalization penalty
- No antisymmetry architecture
- Pure gradient-descent on residual

Variational structure must be built into the loss function.

Ansatz: Slater–Jastrow–Neural Correlator

$$\Psi_{\theta,\beta}(R) = \underbrace{\mathrm{SD}(R')}_{\text{antisymmetry}} \; \exp\Big(\underbrace{U(R)}_{\text{analytic pair}} + \underbrace{W_\theta(R)}_{\text{neural correlator}} \Big), \qquad R' = R + \underbrace{\Delta_\beta(R)}_{\text{backflow}}$$

Four components

- $\mathrm{SD}(R')$ — Slater determinant: exact fermionic antisymmetry
- $U(R)$ — analytic cusp/pair term (evaluated on *unscaled* distances r_{ij})
- $W_\theta(R)$ — compact permutation-invariant neural correlator (evaluated on *trap-scaled* coords $\tilde{\mathbf{x}}_i = \sqrt{\omega}\,\mathbf{x}_i$)
- $\Delta_\beta(R)$ — optional backflow, only inside SD

Coordinate convention

W_θ uses trap-scaled $\tilde{\mathbf{x}}_i = \sqrt{\omega}\,\mathbf{x}_i$; U uses physical r_{ij}.

This separates the length scales set by the trap from those set by the Coulomb interaction.

Two backflow variants:
PINN+BF (`BackflowNet`) and
PINN+CTNN (`CTNNBackflowNet`)

Analytic Pair Term and Backflow Modules

Analytic cusp pair term $U(R)$

$$U(R) = \sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}), \quad u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r}$$

Spin-dependent couplings satisfying cusp conditions:

$$\gamma_{\uparrow\downarrow} = \tfrac{1}{d-1}, \qquad \gamma_{\uparrow\uparrow} = \tfrac{1}{d+1}$$

Evaluated on *unscaled* distances $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$.

Backflow $\Delta_\beta(R)$

Shifts only the Slater determinant: $R' = R + \Delta_\beta(R)$.

`BackflowNet`: pairwise message-passing MLP; output $s_\beta \tanh(\delta\mathbf{x}_i)$.

`CTNNBackflowNet`: explicit node features $\mathbf{h}_i \in \mathbb{R}^{H_v}$ and edge features $\mathbf{h}_{ij} \in \mathbb{R}^{H_e}$; learned linear transports between node and edge spaces.

CTNN Backflow: Standard vs. Learned Coordinate Deformation

Standard backflow

Effective coordinates via an *analytic* pairwise form:

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + \sum_{j\neq i} \eta(r_{ij})(\mathbf{r}_i - \mathbf{r}_j)$$

Physically motivated, few parameters, but limited expressivity and no access to higher-body structure.

CTNN backflow

Replaces the analytic form with a *learned* many-body map:

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + F_{i,\theta}(\mathbf{R})$$

Node features $\mathbf{h}_i$ and edge features $\mathbf{h}_{ij}$ are computed by a graph-neural-network layer; linear transports between node and edge spaces produce the displacement $F_{i,\theta}$.

Feature	Standard BF	CTNN BF
Functional form	analytic	learned
Structure	pairwise	many-body / hierarchical
Parameters	few	many
Expressivity	limited	high

CTNN Backflow: Antisymmetry and the Variational Principle

How the Slater determinant is modified

Without backflow each row of the Slater matrix depends only on particle i: $M_{ia} = \phi_a(\mathbf{r}_i)$.

With CTNN backflow, orbitals are evaluated at *effective* coordinates:

$$\widetilde{M}_{ia} = \phi_a\big(\mathbf{r}_i + F_{i,\theta}(\mathbf{R})\big).$$

Every row now depends on the *full* configuration — these are configuration-dependent quasi-orbitals.

Antisymmetry preserved

If $F_{i,\theta}$ is permutation-equivariant, swapping $i \leftrightarrow j$ swaps two rows of $\widetilde{M}$, so $D_{\mathrm{BF}} \mapsto -D_{\mathrm{BF}}$. ✓

Variational role

CTNN backflow enlarges the variational manifold beyond Jastrow + orbital parameters to include the learned many-body geometry of the effective coordinates, and can improve both many-body correlations and the nodal surface — potentially lowering $E(\theta)$.

Ground-State Energies: Comparison with DMC

N	ω	DMC reference	PINN+BF	% err	PINN+CTNN	% err
2	0.01	$0.073839(2)$	$0.073838(1)$	-0.0014	$0.0738382(5)$	-0.0014
	1.00	3.00000	$2.99999(1)$	-0.0003	$3.00000(2)$	-0.0000
6	0.10	$3.55385(5)$	$3.5549(1)$	$+0.030$	$\mathbf{3.55388(5)}$	$+0.0008$
	0.50	$11.78484(6)$	$11.7895(4)$	$+0.040$	$\mathbf{11.7847(2)}$	-0.0000
	1.00	$20.15932(8)$	$20.1610(6)$	$+0.008$	$\mathbf{20.1585(3)}$	-0.004
12	0.50	$39.1596(1)$	$39.1786(8)$	$+0.049$	$\mathbf{39.1604(3)}$	$+0.002$
	1.00	$65.7001(1)$	$65.717(1)$	$+0.026$	$\mathbf{65.69556(5)}$	-0.007
20	0.10	$29.9779(1)$	—	—	$29.9888(2)$	$+0.036$
	0.50	$93.8752(1)$	—	—	$\mathbf{93.8789(5)}$	$+0.004$
	1.00	$155.8822(1)$	—	—	$\mathbf{155.8738(7)}$	-0.002

- $N = 2$: both ansätze reproduce DMC within statistical error for all ω
- $N \geq 6$: PINN+CTNN performs somewhat better than PINN+BF
- $N = 20$: CTNN remains accurate at scale; errors $\lesssim 0.04\%$

Full Ground-State Energy Table I

N	ω	DMC ref.	PINN+BF	% err	PINN+CTNN	% err
2	10^{-3}	—	0.0137948(8)	—	0.013778(1)	—
	10^{-2}	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0014
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	−0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	−0.0000
6	10^{-3}	—	0.140913(1)	—	0.1408172(7)	—
	10^{-2}	—	0.69125(2)	—	0.69036(1)	—
	0.10	3.55385(5)	3.5549(1)	+0.0295	3.55388(5)	+0.0008
	0.50	11.78484(6)	11.7895(4)	+0.0395	11.7847(2)	−0.0000
	1.00	20.15932(8)	20.1610(6)	+0.0083	20.1585(3)	−0.0041
12	10^{-3}	—	0.515823(3)	—	0.515365(4)	—
	10^{-2}	—	2.48620(5)	—	2.47363(4)	—
	0.10	12.26984(8)	12.2731(2)	+0.0160	12.2718(1)	—
	0.50	39.1596(1)	39.1786(8)	+0.0485	39.1604(3)	+0.0018
	1.00	65.7001(1)	65.717(1)	+0.0257	65.69556(5)	−0.0069
20	10^{-3}	—	—	—	1.293033(6)	—
	10^{-2}	—	—	—	6.14645(5)	—
	0.10	29.9779(1)	—	—	29.9888(2)	+0.0364
	0.50	93.8752(1)	—	—	93.8789(5)	+0.0040
	1.00	155.8822(1)	—	—	155.8738(7)	−0.0024

Energy Partitioning: Approaching the Wigner Regime

Total energy decomposition:

$$E = \langle T\rangle + \langle V_{\mathrm{int}}\rangle + \langle V_{\mathrm{trap}}\rangle$$

Two diagnostics tracking interaction dominance:

$$\Gamma(\omega) = \frac{\langle V_{\mathrm{int}}\rangle}{\langle T\rangle}, \qquad \mathcal{C}(\omega) = \frac{2\langle V_{\mathrm{trap}}\rangle}{\langle V_{\mathrm{int}}\rangle}$$

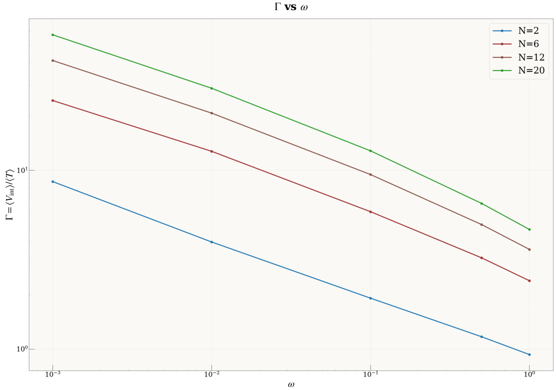
- $\Gamma\uparrow$ as $\omega\downarrow$: interactions dominate
- $\mathcal{C} \to 1$: quasi-classical trap–Coulomb balance

N	$\Gamma_{\omega=1}$	$\mathcal{C}_{\omega=1}$	$\Gamma_{\omega=10^{-3}}$	$\mathcal{C}_{\omega=10^{-3}}$
2	0.93	3.14	8.64	1.226
6	2.41	1.83	24.55	1.074
12	3.61	1.56	41.11	1.049
20	4.66	1.43	57.26	1.038

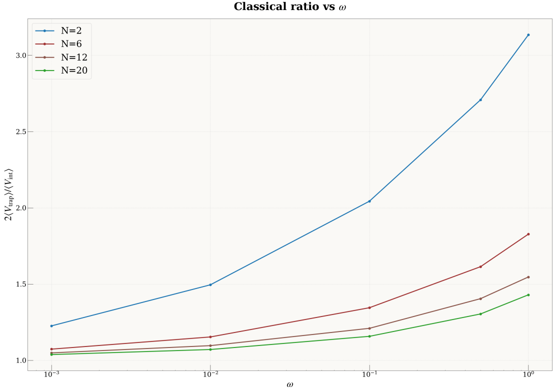
Key trend

At $\omega = 10^{-3}$, $N \geq 12$: $\mathcal{C} < 1.05$, i.e. $V_{\mathrm{int}} \approx 2V_{\mathrm{trap}}$ to $\lesssim 5\%$.

Ratios of Expectation Values



(a) $\Gamma(\omega) = \langle V_{\mathrm{int}}\rangle / \langle T\rangle$ (log–log). Interaction dominance increases as $\omega \to 0$.



(b) $\mathcal{C}(\omega) = 2\langle V_{\mathrm{trap}}\rangle / \langle V_{\mathrm{int}}\rangle$.

Reproducibility Across Seeds: CI Benchmark ($N = 2$–6, $\omega = 1$)

N	E_{CI} (au)	Seed 42	Seed 314	Seed 901	Spread (au)	Max |err|
2	2.254424	2.253853	2.254049	2.253171	0.00088	0.056%
3	3.637002	3.636503	3.636209	3.636661	0.00029	0.022%
4	5.104449	5.103305	5.103369	5.103568	0.00026	0.022%
6	8.217032†	8.213596	8.214344	8.213966	0.00075	0.042%

† $N = 6$: $n_{\mathrm{sp}} = 10$, $n_{\mathrm{ci}} = 500$ (retains 0.05% of 10^6 configurations).

Key result

Every single run lies below its CI reference. Non-MCMC training achieves sub-0.06% accuracy across all N, with three independent seeds. Loss functions do not increase with N (tentative results).

Summary: Energetics

Accuracy with respect to DMC

- Residual pretraining + SR tail pipeline
- $N = 2$: both PINN+BF and PINN+CTNN agree with DMC to within statistical uncertainty
- $N = 6, 12, 20$: PINN+CTNN seems to perform better than PINN+BF at moderate/strong confinement
- Deviations from DMC: $\lesssim 0.05\%$ in benchmarked regimes

Energy partitioning

- $\Gamma(\omega)$: interaction-to-kinetic ratio grows from < 1 to ~ 57 across the ω range
- r_{rms}: $\sim 70\times$ expansion from $\omega = 1$ to $\omega = 10^{-3}$

Methodological Contributions

1. **Architecture.** Permutation-invariant DeepSets + pair features; PINN loss with explicit physical constraints.

Broader applicability

Same PINN framework is directly applicable to: other 2D/3D trapped systems · real-time dynamics · excited states · open quantum systems · device-level quantum dot modelling.

PINNs vs. VMC: Complementary Strengths

PINN advantages

- Wave function is a continuous function of coordinates
- Natural access to excited states, complex geometries, and time-dependent problems
- Antisymmetry and normalisation as modular, explicit constraints

VMC advantages

- Established workhorse for large-N ground states
- Efficient MC sampling of high-dimensional spaces
- Variational upper bounds by construction
- Rich library of optimized ansatz forms

Outlook

VMC excels at large-N ground states; PINNs are strong candidates for continuous-space, dynamical, constrained, and few-to-intermediate-body problems. Cross-fertilization between the two frameworks is an important direction.

Open Questions and Future Directions

Computational / methodological

- Scaling to $N > 20$: sparse/message-passing architectures to reduce $O(N^2)$ cost, work in progress
- Transfer learning: pre-train on one ω, fine-tune across the full ω grid
- Excited states: add orthogonality constraints to loss
- Real-time dynamics: extend PINN loss with time dependence
- Improved optimizers: ?

Physics questions

- Transition to Wigner-Seitz
- Spin degree of freedom and magnetic field effects
- Coupled dots and transport / conductance calculations
- Entanglement structure of the many-body wave function

Conclusions

1. **PINNs are accurate** for quantum dots at $N = 2$–20 across five orders of magnitude in ω, matching DMC benchmarks to $\lesssim 0.05\%$.
2. **Energy partitioning and virial validation** confirm physical correctness even without DMC references.
3. **PINNs provide a unified, physics-grounded framework** for few-to-intermediate quantum many-body problems in continuous space — with natural extensions to dynamics and excited states.

CTNN Backflow vs. Standard Backflow: Aim

These slides provide the full derivation supporting the two summary slides in the main talk. They cover:

1. What standard backflow is and its limitations.
2. How CTNN backflow replaces the analytic form with a learned many-body map.
3. How backflow modifies the Slater determinant mathematically.
4. How the full construction enters the variational principle.

The main physical context is a fermionic many-body wave function where one wants a flexible yet antisymmetric ansatz that captures strong correlations and improves the nodal surface.

Ordinary Slater–Jastrow Ansatz

A common fermionic variational ansatz is

$$\Psi(\mathbf{R}) = e^{J(\mathbf{R})} \det[\phi_a(\mathbf{r}_i)], \qquad \mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N).$$

- $J(\mathbf{R})$ is a symmetric Jastrow factor capturing dynamic correlations.
- The determinant enforces fermionic antisymmetry.
- The single-particle orbitals are evaluated at the *bare* coordinates $\mathbf{r}_i$.

Limitation

Jastrow factors alone cannot move the nodal surface of Ψ, which determines the fixed-node error in DMC and the quality of the variational bound. Backflow addresses this by modifying the coordinates *before* they enter the determinant.

Standard Backflow

Standard backflow introduces *effective* (quasi-particle) coordinates

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + \boldsymbol{\xi}_i(\mathbf{R}),$$

with the classic pairwise ansatz

$$\boldsymbol{\xi}_i(\mathbf{R}) = \sum_{j\neq i} \eta(r_{ij})\,(\mathbf{r}_i - \mathbf{r}_j), \qquad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|.$$

Particle i is displaced by a weighted sum of pairwise vectors from all other particles; $\eta(r)$ is a short-ranged scalar function.

Strengths

- Physically motivated
- Analytically tractable
- Easy to interpret

Limitations

- Restricted functional flexibility
- Cannot capture higher-body structure
- Tied to a chosen analytic form

CTNN Backflow: Core Idea

CTNN backflow replaces the fixed analytic pairwise form by a *learned* many-body map:

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + F_{i,\theta}(\mathbf{R}), \qquad \text{or collectively,} \qquad \tilde{\mathbf{R}} = F_\theta(\mathbf{R}).$$

The essential conceptual shift is

$$\underbrace{\text{analytic pairwise deformation}}_{\text{standard BF}} \longrightarrow \underbrace{\text{learned many-body coordinate transformation}}_{\text{CTNN BF}}.$$

Generic CTNN construction

$$h_i = \phi_\theta(\mathbf{r}_i), \qquad h_{ij} = \psi_\theta(\mathbf{r}_i, \mathbf{r}_j), \qquad H_i = \sum_{j \neq i} h_{ij},$$
$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + g_\theta(h_i, H_i).$$

The functions ϕ_θ, ψ_θ, g_θ are learned and can represent non-trivial collective structure beyond pairwise corrections.

Standard vs. CTNN Backflow: Feature Comparison

Feature	**Standard BF**	**CTNN BF**
Functional form	analytic	learned
Typical structure	pairwise	many-body / hierarchical
Parameters	few	many
Expressivity	limited	high
Interpretability	high	moderate

$\boxed{\text{Standard BF} = \text{analytic pairwise correction}}$ $\boxed{\text{CTNN BF} = \text{learned many-body deformation}}$

Backflow-Modified Slater Determinant

Without backflow

$$M_{ia}(\mathbf{R}) = \phi_a(\mathbf{r}_i), \quad D(\mathbf{R}) = \det M(\mathbf{R}).$$

Each row depends *only* on the coordinate of one particle.

With backflow

$$\widetilde{M}_{ia}(\mathbf{R}) = \phi_a\big(\tilde{\mathbf{r}}_i(\mathbf{R})\big),$$

$$D_{\mathrm{BF}}(\mathbf{R}) = \det\big[\phi_a(\tilde{\mathbf{r}}_i(\mathbf{R}))\big].$$

Each row now depends on the *full* configuration through $\tilde{\mathbf{r}}_i(\mathbf{R})$.

What has changed mathematically?

After backflow, $\widetilde{M}_{ia} = \phi_a(\mathbf{r}_i + F_{i,\theta}(\mathbf{R}))$, so every row becomes a many-body object — yet the determinant remains antisymmetric (see next slide). The orbitals become *configuration-dependent quasi-orbitals*.

Why Antisymmetry Is Preserved

Permutation equivariance

If the backflow map $F_{i,\theta}$ is permutation-equivariant, then exchanging particles $i \leftrightarrow j$ exchanges the corresponding effective coordinates:

$$\tilde{\mathbf{r}}_i(\mathbf{R}) \longleftrightarrow \tilde{\mathbf{r}}_j(\mathbf{R}).$$

Hence two rows of $\widetilde{M}$ are exchanged, giving

$$D_{\mathrm{BF}}(\mathbf{R}) \mapsto -D_{\mathrm{BF}}(\mathbf{R}).$$

Antisymmetry is preserved. ✓

Small-displacement expansion

For small backflow $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \delta\mathbf{r}_i$:

$$\widetilde{M}_{ia} \approx \phi_a(\mathbf{r}_i) + F_{i,\theta}(\mathbf{R}) \cdot \nabla\phi_a(\mathbf{r}_i) + \cdots$$

Backflow introduces configuration-dependent gradient corrections inside the determinant.

Full Ansatz: Backflow Plus Jastrow

The complete variational wave function is

$$\Psi_\theta(\mathbf{R}) = e^{J_\theta(\mathbf{R})} \det\big[\phi_a\big(\mathbf{r}_i + F_{i,\theta}(\mathbf{R})\big)\big].$$

- J_θ handles *symmetric* (Jastrow) correlations.
- The determinant enforces fermionic antisymmetry.
- CTNN backflow changes the *nodal structure* through the effective coordinates — something Jastrow factors alone cannot do.

Key point

CTNN backflow is not a preprocessing step: it is an intrinsic part of the variational wave function, and its parameters θ are optimized together with the Jastrow and orbital parameters.

CTNN Backflow and the Variational Principle

The variational energy is

$$E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} \geq E_0.$$

With CTNN backflow the parameter set θ includes the learned map $F_{i,\theta}$, so the *variational manifold is enlarged* beyond Jastrow + orbital coefficients to include the many-body geometry of the effective coordinates.

Two ways backflow helps

1. Captures additional many-body correlations
2. Improves the nodal surface of Ψ_θ

Both effects can lower $E(\theta)$ toward E_0.

Derivative structure

Spatial derivatives involve a chain rule through $\tilde{\mathbf{r}}_i(\mathbf{R})$, modifying gradients, Laplacians, the kinetic energy, and parameter gradients for optimization.

Full Variational Optimization Problem

The optimization target is

$$\theta^\star = \arg\min_\theta \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle}, \qquad \Psi_\theta(\mathbf{R}) = e^{J_\theta(\mathbf{R})} \det\big[\phi_a\big(\mathbf{r}_i + F_{i,\theta}(\mathbf{R})\big)\big].$$

Local energy in VMC / PINN+SR

$$E_{\mathrm{loc}}(\mathbf{R}) = \frac{H\Psi_\theta(\mathbf{R})}{\Psi_\theta(\mathbf{R})}, \qquad E(\theta) = \frac{\int |\Psi_\theta|^2 E_{\mathrm{loc}}\, d\mathbf{R}}{\int |\Psi_\theta|^2\, d\mathbf{R}}.$$

Optimization requires evaluating the backflow-modified determinant and its derivatives on sampled configurations at each step.

In the PINN+CTNN pipeline, the SR tail phase minimizes $E(\theta)$ directly, restoring the variational upper-bound property after residual-based pretraining.

CTNN Backflow: Summary

- Standard backflow uses an analytic pairwise correction $\boldsymbol{\xi}_i = \sum_{j\neq i} \eta(r_{ij})(\mathbf{r}_i - \mathbf{r}_j)$.
- CTNN backflow uses a learned many-body map $F_{i,\theta}(\mathbf{R})$ built from node and edge features in a graph-neural-network layer.
- The Slater determinant is modified by evaluating orbitals at effective coordinates: $\phi_a(\mathbf{r}_i) \;\to\; \phi_a(\tilde{\mathbf{r}}_i(\mathbf{R}))$.
- Antisymmetry is preserved if the map is permutation-equivariant.
- CTNN backflow enlarges the variational manifold and can lower the ground-state energy by improving many-body correlations and the nodal surface.

Take-home formulas

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i + F_{i,\theta}(\mathbf{R}), \qquad \Psi_\theta = e^{J_\theta} \det[\phi_a(\tilde{\mathbf{r}}_i)], \qquad E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle}.$$

Activation Functions and Higher-Order Derivatives

- PINNs require second- (or higher-) order derivatives through AD
- Activation function choice is critical for $\ell \geq 2$ PDEs
- **Smooth activations required:** $\tanh$, SiLU, Softplus
- **ReLU is unsuitable** for second-order operators (second derivative is zero almost everywhere)

Network Architecture

Ground-state PINN Jastrow

- 2 hidden layers, width 64, GELU activations
- Input: all pairwise distances + absolute positions
- Output: scalar log-amplitude correction J
- Backflow: optional (2 layers, width 32); disabled in all production runs
- Parameters: $\sim$25K ($N = 2$) to $\sim$45K ($N = 6$)

Spectral network (imaginary-time evolution)

- 3 temporal modes $\{A_k(\tau)\}$ + constant
- Spatial network: 2 layers, width 32 nodes
- Boundary condition: $g \to 0$ as $\tau \to \infty$ enforced analytically

Wavefunction form

$\psi = \exp(J) \cdot D$ (Jastrow × Slater determinant)

Singlet:

$\log |\psi_{\text{singlet}}| = \text{logaddexp}(J(\mathbf{r}_1, \mathbf{r}_2),\ J(\mathbf{r}_2, \mathbf{r}_1))$

Training Protocol

Ground-state training (all production runs)

- Optimizer: Adam, lr $= 10^{-4}$, cosine warm-up (200–400 epochs), $\text{lr}_{\min} = 0.1 \times$ lr
- Batch size n_{coll}: 512 ($N \leq 4$), 1024 ($N = 6$), 2048 ($N = 8$)
- Laplacian: finite difference ($h = 0.01$) for $N > 4$
- Gradient clip: 0.2

Training Pipeline and Benchmark Strategy

Training pipeline

1. **Residual-based pretraining:** PINN loss enforces the many-body Schrödinger equation at collocation points
2. **Stochastic Reconfiguration (SR) tail:** second-order-like variational refinement; improves energy and wave-function quality

Internal consistency check

Harmonic-plus-Coulomb virial identity:

$$2\langle T\rangle = 2\langle V_{\text{trap}}\rangle - \langle V_{\text{int}}\rangle$$

Satisfied to sub-percent accuracy (worst case $\lesssim 0.4\%$).

Benchmark scope

- $N = 2, 6, 12, 20$ electrons
- $\omega \in \{10^{-3}, 10^{-2}, 0.1, 0.5, 1.0\}$
- Compared to diffusion Monte Carlo (DMC) where available
- Two ansätze: PINN+BF and PINN+CTNN

Exact Diagonalization Reference

CI via 2D DVR (discrete variable representation)

1. Each of K wells contributes $n_{\rm sp}$ single-particle orbitals from its local HO basis
2. N-particle product space: $n_{\rm sp}^N$ configurations, sorted by non-interacting energy

N	$n_{\rm sp}$	$n_{\rm ci}$	grid	SP energy
2–4	40	300	24×24	0.998
6	10	500	24×16	1.000